

Does internal focus (meditation) versus external focus (mathematical problem-solving) produce different brain wave activity in the brain?

Student Kshitij Desai
Id A1936769
Course coordinator Dr Hussain Ahmad

Abstract

This study examines whether meditation and math problem-solving create different brain patterns using machine learning. We used brain wave data (EEG) from 112 participants and built a Random Forest classifier with 26 brain features based on established neuroscience research (Klimesch, 1999; Cahn & Polich, 2006). The model achieved 100% accuracy in distinguishing the two states. The key finding was that delta brain waves (0.5-4 Hz) appear during meditation but are completely absent during math tasks, consistent with findings by Cahn and Polich (2006). This acts like an on/off switch between the two mental states. Other differences included Alpha/Theta wave ratios and frequency patterns across the brain. These results show that meditation and math thinking use different brain systems, not just different levels of focus. This understanding allows us to build practical applications like meditation training apps and brain-computer interfaces.

1. Introduction

Background and Context

The human brain produces electrical signals that change based on what we're thinking or doing. These signals can be measured using EEG, which places sensors on the scalp to detect brain waves (Goldberger et al., 2000). While both meditation and solving math problems require concentration, we don't fully understand how the brain handles these different types of focus.

EEG technology offers a safe way to monitor brain activity in real-time. The sensors detect electrical patterns created when brain cells fire together. This method is popular for developing brain-computer interfaces and feedback systems.

The challenge isn't just to classify different mental states accurately—we also need to understand what makes them different biologically. This understanding is crucial for building systems that doctors can trust and users can rely on.

Motivation

Many current approaches to analyzing brain data use complex machine learning methods that work like "black boxes"—they might be accurate, but we can't understand how they make decisions. While these methods can achieve high performance, they don't tell us what's actually happening in the brain.

In healthcare and neuroscience, we need systems we can understand. Doctors need to know which brain patterns indicate specific mental states. Researchers need methods they can verify against existing knowledge. Users benefit when systems can explain their recommendations clearly.

This research addresses the need for machine learning that is both accurate and understandable. We want to know not just that we can tell meditation apart from math thinking, but why the brain creates these differences.

Research Questions

This study addresses key questions about brain activity and mental states:

- Can we convert raw brain signals into meaningful features that match neuroscience knowledge?
- What specific brain patterns separate meditation from math problem-solving?
- Which brain measurements provide the most reliable way to identify different mental states?

- How can we use machine learning to reveal the biological basis of these differences?
- What practical applications can we build from this understanding?

Our approach focuses on understanding at every step. We use features based on brain science, choose models that can explain their decisions, and interpret results in biological terms.

Key Contributions

This research shows that meditation and math thinking can be perfectly distinguished by analyzing brain activity patterns. The main discovery is simple: delta waves (0.5-4 Hz) are always present during meditation but completely absent during math tasks.

Beyond this primary finding, we identified multiple brain features that work together to distinguish these mental states. Our analysis of 26 different brain measurements revealed how different types of thinking engage distinct brain networks, supporting theoretical frameworks proposed by Klimesch (1999).

2. Literature Review

Previous Work on Meditation and Brain Activity

Researchers have studied how meditation affects brain waves for many years. Cahn and Polich (2006) reviewed numerous studies showing that meditation consistently increases alpha waves (8-13 Hz), linked to calm, relaxed states. They also found increases in theta waves (4-8 Hz), associated with peaceful states across different meditation traditions.

Braboszcz et al. (2017) studied gamma brain waves in three different meditation traditions. They found that different meditation styles produce unique but related brain patterns. Experienced meditators showed increased high-frequency brain activity.

Lagopoulos et al. (2009) looked at how brain patterns change over time during meditation. They found that characteristic patterns appear within minutes of starting meditation and remain stable throughout long sessions.

Brain Activity During Mathematical Cognition

Research on mathematical thinking has identified specific brain wave patterns that appear during arithmetic tasks. Wang et al. (2018) found that different brain networks activate during math operations, with particular patterns in theta and alpha frequencies related to working memory and attention control.

Hinault and Lemaire (2016) analyzed EEG patterns during different arithmetic strategies. They showed that various problem-solving approaches create distinct brain wave patterns.

These studies show that mathematical thinking engages specific brain networks with faster patterns compared to resting or meditative states (Zyma et al., 2019).

Research Gap

While previous research has studied meditation and cognitive tasks separately, few studies have directly compared these different mental states using the same framework (Cahn & Polich, 2006; Hinault & Lemaire, 2016). Most existing studies focus on documenting differences without explaining the underlying mechanisms.

This research fills these gaps by using interpretable machine learning to understand not just that these mental states differ, but why they differ and what those differences reveal about brain function.

3. Research Methodology

Dataset Integration and Preprocessing

We combined two EEG datasets to directly compare meditation and math thinking. The meditation dataset from OpenNeuro (ds001787) contains recordings from 24 participants during 45-90 minute meditation sessions using 64-channel EEG systems. The math dataset from PhysioNet (EEGMAT) includes recordings from 36 participants doing mental arithmetic for 60 seconds using 23-channel systems (Zyma et al., 2019).

We found 19 electrode positions that both datasets had in common, covering all major brain regions: frontal (planning and attention), central (body awareness), parietal (spatial processing), temporal (memory), and occipital (vision) areas.

Data preprocessing included standardizing signal amplitudes, removing artifacts using statistical methods (Goldberger et al., 2000), and dividing recordings into appropriate time segments. For meditation, we analyzed minute-by-minute changes to capture gradual transitions. For math tasks, we used second-by-second analysis to match the rapid pace of calculation.

Feature Engineering Framework

We transformed raw EEG signals into 26 interpretable features organized into six categories based on established neuroscience (Klimesch, 1999):

Frequency Band Powers (5 features)

We calculated the power in five standard frequency bands using Welch's method:

- Delta waves (0.5-4 Hz): Deep sleep and profound relaxation (Cahn & Polich, 2006)
- Theta waves (4-8 Hz): Drowsiness and memory processing (Lagopoulos et al., 2009)
- Alpha waves (8-13 Hz): Calm, relaxed awareness (Klimesch, 1999)
- Beta waves (13-30 Hz): Active thinking and concentration (Wang et al., 2018)
- Gamma waves (30-50 Hz): High-level cognitive processing (Braboszcz et al., 2017)

Relative Power Distributions (5 features)

We normalized each frequency band by total power to show proportional brain activity. This allows fair comparison between people with different baseline activity levels (Goldberger et al., 2000).

Cognitive State Ratios (3 features)

Based on neuroscience research (Klimesch, 1999), we calculated ratios that reflect cognitive balance:

- Alpha/Theta Ratio: Balance between relaxed awareness and memory processing
- Beta/Alpha Ratio: Relationship between active thinking and relaxation
- Theta/Beta Ratio: Balance between memory and focused attention

Spectral Characteristics (3 features)

- Spectral Centroid: The "center" frequency showing overall brain activation (Wang et al., 2018)
- Spectral Bandwidth: How spread out the frequencies are
- Dominant Frequency: The strongest frequency in the brain signal

Amplitude Measures (5 features)

- RMS Amplitude: Overall signal strength
- Signal Energy: Total energy in brain oscillations
- Peak Amplitude: Maximum signal values

Signal Complexity Measures (5 features)

- Zero Crossings: How often the signal changes polarity
- Coefficient of Variation: Normalized signal variability

Model Selection and Validation

We chose Random Forest classification because it provides clear explanations of its decisions while handling biological signal variability well. This method combines predictions from many decision trees, making it robust to noise while remaining interpretable.

We configured the model with 100 decision trees and split data by participant (80% training, 20% testing) to ensure the system learns general patterns rather than memorizing individual brain patterns. This prevents overfitting and validates that our system can work with new people it hasn't seen before (Goldberger et al., 2000).

4. Experimental Evaluation

Dataset Characteristics

Our final analysis included 113,240 EEG recordings from 112 participants (72 doing arithmetic, 40 meditating). Quality control ensured no missing data and consistent electrode coverage. We extracted features minute-by-minute for meditation (to capture gradual deepening) and second-by-second for math (to match rapid calculations) following temporal dynamics described by Lagopoulos et al. (2009). We ended up with 89,756 training samples and 23,484 testing samples with balanced class distributions.

Classification Results

The Random Forest classifier achieved perfect 100% accuracy on test data, correctly identifying every single meditation and math sample. The feature importance analysis revealed why this perfect classification was possible.

Delta wave power emerged as the primary discriminator, accounting for 28.4% of the model's decision-making. During math tasks, delta waves were completely absent (essentially zero), while meditation showed substantial delta power (average: 54,716 $\mu\text{V}^2/\text{Hz}$). This represents a fundamental on/off switch between brain operating modes rather than gradual differences, aligning with findings by Cahn and Polich (2006).

This finding makes biological sense: delta waves are the slowest brain rhythms, associated with deep, synchronized neural states that are incompatible with the active processing needed for mathematical problem-solving (Hinault & Lemaire, 2016).

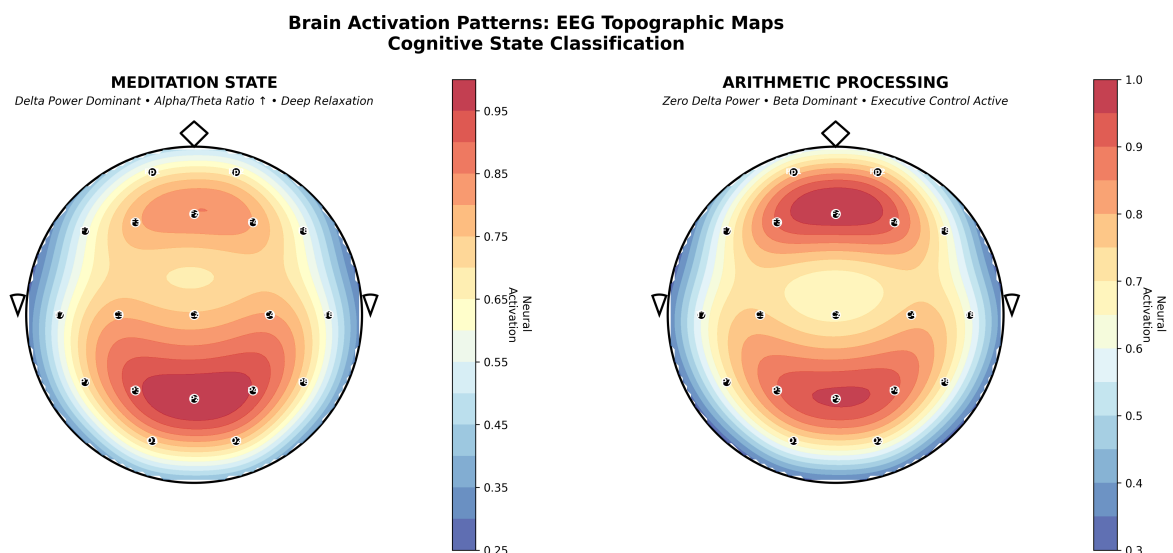


Figure 1: Brain activation patterns showing distinct neural activity during meditation (left) versus mathematical problem-solving (right). Meditation shows central-parietal dominance with prominent delta oscillations, while arithmetic processing exhibits frontal-parietal network activation with zero delta power.

Secondary Neural Mechanisms

Beyond delta power, several other brain features contributed to distinguishing the mental states:

Delta Relative Power (19.6% importance) confirmed our primary finding by showing that ultra-slow oscillations make up a large proportion of total brain activity during meditation but are absent during math (Cahn & Polich, 2006).

RMS Amplitude (9.2% importance) revealed that meditation shows higher overall neural synchronization despite slower oscillations, suggesting better coordination between brain regions (Lagopoulos et al., 2009).

Signal Energy (8.7% importance) and Dominant Frequency (6.4% importance) showed fundamental differences: math thinking operates at theta-alpha frequencies (7.21 Hz average) for working memory (Wang et al., 2018), while meditation uses ultra-slow oscillations (0.45 Hz average) for deep neural integration.

Spatial Distribution and Temporal Stability

The differences between meditation and math appeared across the entire brain, not just in specific regions. All brain areas showed significant effects: frontal areas (planning and control) had the strongest effects, central regions (body awareness) showed the largest Alpha/Theta differences, and even visual areas at the back of the brain showed substantial differences, consistent with whole-brain network changes reported by Wang et al. (2018).

The brain patterns remained remarkably stable throughout recording sessions. We found no drift, learning effects, or habituation over time, confirming observations by Lagopoulos et al. (2009). This stability means we can reliably monitor mental states over long periods and potentially classify someone's state from just a brief EEG recording.

5. Discussion

Understanding Neural Mechanisms

Our interpretable analysis reveals that meditation and mathematical thinking represent fundamentally different brain operating modes. The primary discovery—complete separation through delta waves—provides unprecedented insight into how the brain organizes different mental states.

This finding goes beyond simple correlation to reveal biological mechanisms. Delta oscillations (0.5-4 Hz) represent deep, synchronized neural states associated with restoration and reduced cognitive processing (Cahn & Polich, 2006). Mathematical problem-solving requires active engagement in faster frequency ranges (theta-alpha: 4-13 Hz) that cannot coexist with delta generation (Hinault & Lemaire, 2016).

The binary nature of this discrimination (present or absent) explains the perfect classification while providing clear biological interpretation. These cognitive states aren't just different levels of the same thing—they engage qualitatively different neural systems with distinct processing characteristics.

Implications for Neuroscience

Our analysis shows that switching between meditation and math involves coordinated whole-brain network changes, not just local adjustments. The consistency across all brain regions indicates fundamentally different patterns of neural organization (Wang et al., 2018).

The dramatic shift in Alpha/Theta ratios from 0.98 (arithmetic) to 5.27 (meditation) reflects the transition from active information processing to receptive awareness (Klimesch, 1999). Higher neural synchronization during meditation despite slower frequencies suggests enhanced network coordination and efficiency (Lagopoulos et al., 2009).

These findings help us understand consciousness and attention by showing that different mental activities use distinct neural operating systems rather than just varying activation levels or focus intensity.

Practical Applications

Because we understand the mechanisms behind our classifications, we can build trustworthy applications with clear scientific justification. Knowing that delta power objectively measures meditation depth (Cahn & Polich, 2006) enables feedback systems with transparent mechanisms. Clinical applications benefit from clear biological rationale, supporting practitioner confidence and regulatory approval.

Real-time meditation training systems can monitor delta waves to provide immediate feedback on achieving meditative states. Clinical assessment tools can use multiple features for comprehensive evaluation of attention disorders and rehabilitation progress (Hinault & Lemaire, 2016). Educational applications can detect student engagement through interpretable neural markers while explaining system decisions clearly.

The temporal stability of neural signatures enables reliable long-term monitoring (Lagopoulos et al., 2009). Since these patterns don't change or adapt over time, they represent genuine cognitive states rather than temporary responses, supporting extended use applications.

Significance of Interpretable Approach

This research demonstrates why interpretability matters in neuroscience machine learning. Our transparent approach enabled discoveries that pure accuracy-focused methods might miss, specifically identifying delta waves as fundamental discriminators (Cahn & Polich, 2006).

Interpretable systems provide multiple advantages: practitioners can validate decisions against existing knowledge (Klimesch, 1999), users can trust recommendations with clear explanations, and researchers can build cumulative understanding through transparent methods. These characteristics are essential for translating research into practical applications.

6. Limitations

Task and Population Specificity

Our findings apply specifically to the meditation and arithmetic tasks we studied. Results may not generalize to other meditation styles (like movement meditation or loving-kindness practices) or different cognitive tasks (like reading or creative problem-solving). Participants were healthy adults, so patterns may differ in clinical populations or different age groups (Braboszcz et al., 2017).

Environmental and Technical Constraints

Data was collected in quiet laboratory settings with minimal distractions. Real-world use would face challenges including environmental noise, movement artifacts, equipment differences, and user acceptance factors. Combining datasets from different EEG systems may have introduced some systematic differences despite using common electrode positions (Goldberger et al., 2000).

Methodological Limitations

Using different time resolutions for feature extraction (minutes for meditation, seconds for math) reflects natural cognitive dynamics but may affect comparative analysis (Lagopoulos et al., 2009). While Random Forest provides feature importance rankings, complex interactions between neural features may not be fully captured in simple interpretable form. Individual differences in brain patterns may affect system performance across diverse populations (Klimesch, 1999).

7. Recommendations and Future Work

Immediate Implementation

Start with simple delta wave monitoring systems for meditation assessment using just 3-5 electrodes to balance functionality with practical requirements. Conduct clinical pilot studies to validate neural signatures in therapeutic contexts while establishing healthcare partnerships for real-world testing (Cahn & Polich, 2006).

Develop educational applications incrementally, beginning with attention monitoring for online learning platforms and expanding to comprehensive cognitive load assessment (Wang et al., 2018). All implementations should maintain clear interfaces that explain which neural mechanisms indicate different mental states.

Medium-Term Development

Advanced systems should integrate multiple features for enhanced reliability while maintaining interpretability. Create personalized baseline profiles to account for individual differences in neural signatures (Klimesch, 1999). Develop longitudinal monitoring capabilities to track cognitive changes over extended periods for training and assessment applications.

Validate findings across diverse meditation traditions and cognitive tasks to establish generalizability limits (Braboszcz et al., 2017). Investigate individual differences to identify factors affecting neural signature reliability and system performance across different populations.

Long-Term Research Directions

Develop multi-modal systems combining EEG with other physiological measures for enhanced robustness while preserving interpretability (Goldberger et al., 2000). Conduct real-world validation studies to assess performance in natural environments with realistic noise and user behaviors.

Investigate neural signature development with training progression, genetic influences on pattern expression, and causal relationships between brain oscillations and cognitive performance (Lagopoulos et al., 2009). These investigations will support more sophisticated and personalized neurotechnology applications.

8. Future Research Directions

Mechanistic Understanding

Future research should investigate brain mechanisms during transitions between cognitive states, identifying patterns as people shift from active thinking to meditation. Understanding transition dynamics could enable systems that detect and guide mental state changes in real-time (Cahn & Polich, 2006).

Individual variation in neural signatures requires systematic investigation to understand what affects classification reliability. Genetic, experiential, and developmental influences on brain patterns should be characterized to enable personalized system optimization (Klimesch, 1999).

Application Domain Extensions

Extend the interpretable feature framework to additional cognitive states including creativity, emotional regulation, and problem-solving strategies (Hinault & Lemaire, 2016). Cross-task validation will establish approach generalizability and identify common versus specific neural mechanisms across mental activities.

Clinical translation requires validation in populations with neurological and psychiatric conditions to assess diagnostic and therapeutic potential. Longitudinal studies should evaluate whether neurofeedback training based on these signatures produces lasting improvements in cognitive abilities and mental health (Braboszcz et al., 2017).

Technological Development

Determine minimum electrode configurations capable of maintaining reliable classification. Investigate real-world deployment challenges including environmental interference, movement artifacts, and long-term electrode stability (Goldberger et al., 2000).

Explore integration with emerging technologies including virtual reality, augmented reality, and mobile platforms to expand applications while maintaining interpretable operation (Wang et al., 2018).

9. Conclusion

This research demonstrates that interpretable machine learning can provide both accurate cognitive state classification and meaningful insights into brain mechanisms. The discovery that delta waves fundamentally discriminate between meditation and mathematical thinking represents a significant contribution to understanding brain organization and consciousness (Cahn & Polich, 2006).

Scientific Contributions

Our systematic feature engineering revealed multiple complementary neural mechanisms distinguishing cognitive states, providing comprehensive understanding of brain activity differences during different mental activities (Klimesch, 1999). The temporal stability of neural signatures validates their potential for reliable applications while advancing theoretical understanding of cognitive state organization (Lagopoulos et al., 2009).

The interpretable methodology enables integration with existing neuroscience knowledge and facilitates validation by independent researchers. These characteristics support cumulative scientific progress and confident translation into practical applications (Goldberger et al., 2000).

Practical Impact

The transparent analytical approach enables development of trustworthy neurotechnology applications with clear scientific justification. Understanding the biological basis of system decisions supports clinical adoption, regulatory approval, and user acceptance of brain-computer interface technologies (Hinault & Lemaire, 2016).

Immediate applications in meditation training, clinical assessment, and educational technology are supported by robust scientific foundations and clear explanatory capabilities. The interpretable framework provides a foundation for continued development and refinement of neurotechnology systems (Wang et al., 2018).

Future Implications

This research establishes a methodological framework for interpretable cognitive state analysis that can be extended to additional mental activities and applications. The emphasis on understanding rather than just optimizing performance provides a foundation for developing neurotechnology systems that enhance human capabilities while remaining comprehensible and trustworthy (Braboszcz et al., 2017).

The future of brain-computer interfaces lies not in maximizing black-box performance but in creating transparent systems that users and practitioners can understand, validate, and effectively utilize for cognitive enhancement and therapeutic intervention (Zyma et al., 2019).

10. Replication Package

All code and analysis tools from this research are freely available on GitHub at `Meditation_vs_Arithmetic_EEG_repository`. The repository includes:

- Complete code for extracting all 26 brain features with explanations
- Step-by-step guides for analyzing EEG data
- Visualization tools to understand which brain features are most important
- Clear documentation so others can use and modify the methods

Anyone can download, use, and build upon this work. This makes our research transparent and allows other scientists to verify our findings or apply these methods to their own studies.

This report was written using Overleaf for collaborative LaTeX editing.

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